

Current NEPA authority, made auditable

The system converts a workbook-controlled Forest Service source library into a validated 3D knowledge graph. It shows what current authority was considered, what applies to a specific Environmental Assessment, what does not apply, and exactly which source evidence supports each compliance finding.

1,916

validated graph nodes

3,442

validated graph edges

33

applicable authorities

340

non-applicable authorities

Current NEPA authority environment mapped as evidence

The graph treats statutes, agency procedures, source records, and review decisions as linked evidence - not as unsupported conclusions.

NEPA statute

Baseline procedural law still governs federal environmental review.



CEQ rulemaking status

CEQ records removal of its NEPA regulations; older 40 CFR 1500-1508 rules are historical reference.



Forest Service layer

USDA Forest Service procedures, regulations, directives, and handbook sources



Reviewer graph

1,916 nodes / 3,442 edges with validation, provenance, and readiness controls.

Why this matters for a client demo

The current regulatory picture is moving from a single CEQ code-of-federal-regulations anchor toward agency-specific implementation. The system shows exactly which source record, directive, forest-plan component, applicability decision, and **Validation passed: 75 graph checks; 33 applicable and 340 non-applicable authority decisions.**

Regulatory-currentness model

The graph is built for a moving NEPA environment. It treats CEQ rulemaking status, Forest Service procedures, source records, directives, forest-plan components, applicability decisions, and readiness blockers as separate nodes instead of flattening them into a narrative answer.

Defensibility model

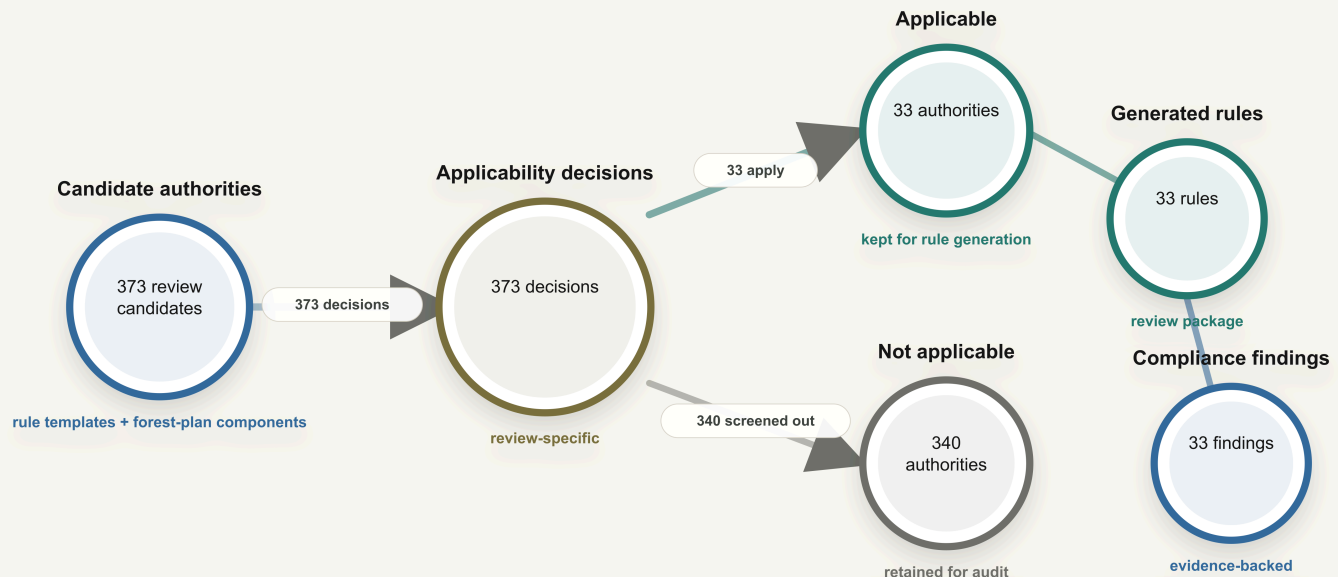
Each viewer claim is backed by graph-export validation. The V1 review overlay passed 75 graph checks and links generated rules and compliance findings back to source records, evidence spans, and review-specific applicability decisions.

From authority universe to review-specific decisions

The graph separates the universe of candidate authorities from the subset that applies to the project. That makes the review politically legible: reviewers can see both what was used and what was considered but screened out.

Applicability is a graph decision layer

The system keeps every candidate authority visible, then separates applicable, non-applicable, generated-rule, and finding nodes for the selected EA review.



Client value shown

- Defensibility: non-applicable authorities are preserved, so reviewers can explain what was considered and why it was screened out.
- Political legibility: the split between authority universe, review decision, generated rule, and compliance finding is visible as graph structure.
- Validation: 75 graph checks passed before this view was exported.

Applicability scene: 373 candidate authority decisions, 33 applicable, 340 non-applicable. The image preserves the graph distinction between candidate authority, review decision, generated rule, and compliance finding.

Defensibility shown

Non-applicable authorities stay in the graph, so the reviewer can explain what was considered and screened out instead of only showing final inclusions.

Political legibility shown

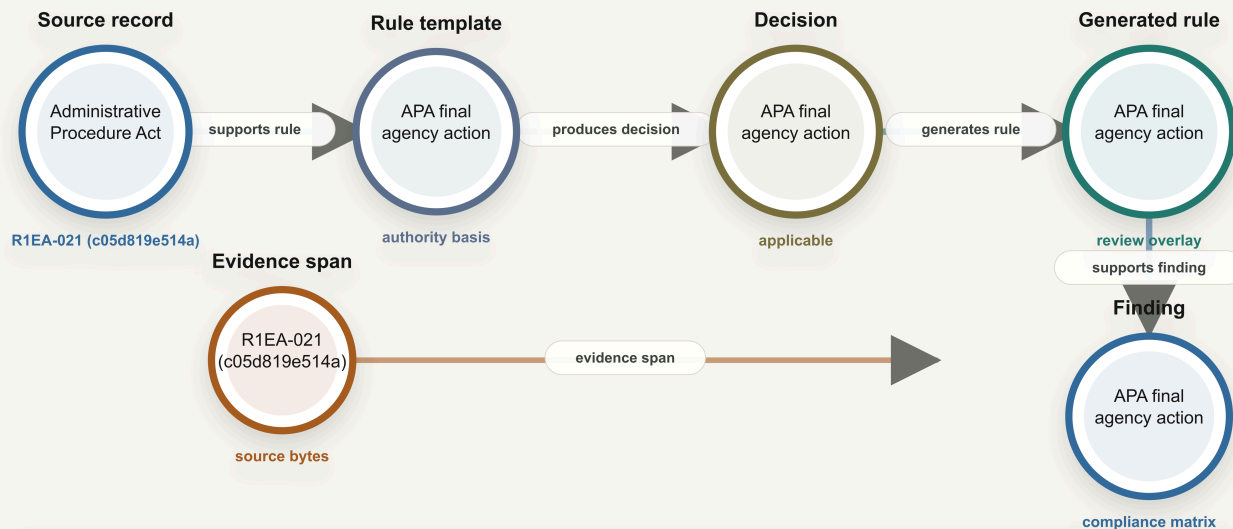
The client sees a visible path from authority universe to decision, rule generation, and compliance finding rather than a black-box conclusion.

From source record to compliance finding

A client should be able to ask "why does this finding exist?" and see the support path. The evidence path view turns that question into a graph traversal from source authority and evidence span through applicability and generated rule support.

A defensible finding is a clickable path

This image is generated from one actual graph trace: source authority, rule template, applicability decision, generated rule, evidence span, and compliance finding.



What the client can inspect

- Authority family: Administrative review, objection/appeal, and final agency action
- Source citation: R1EA-021 (c05d819e514a)
- 99 support edges link generated rules or evidence spans to compliance findings across the V1 overlay.

Evidence path scene: one actual source-record -> rule-template -> applicability-decision -> generated-rule -> compliance-finding path is derived from graph edges, with the evidence span kept visible.

What is inspectable

Source records, citation labels, extraction spans, applicability decisions, generated rules, and findings all remain separate graph objects with explicit support edges.

What this proves

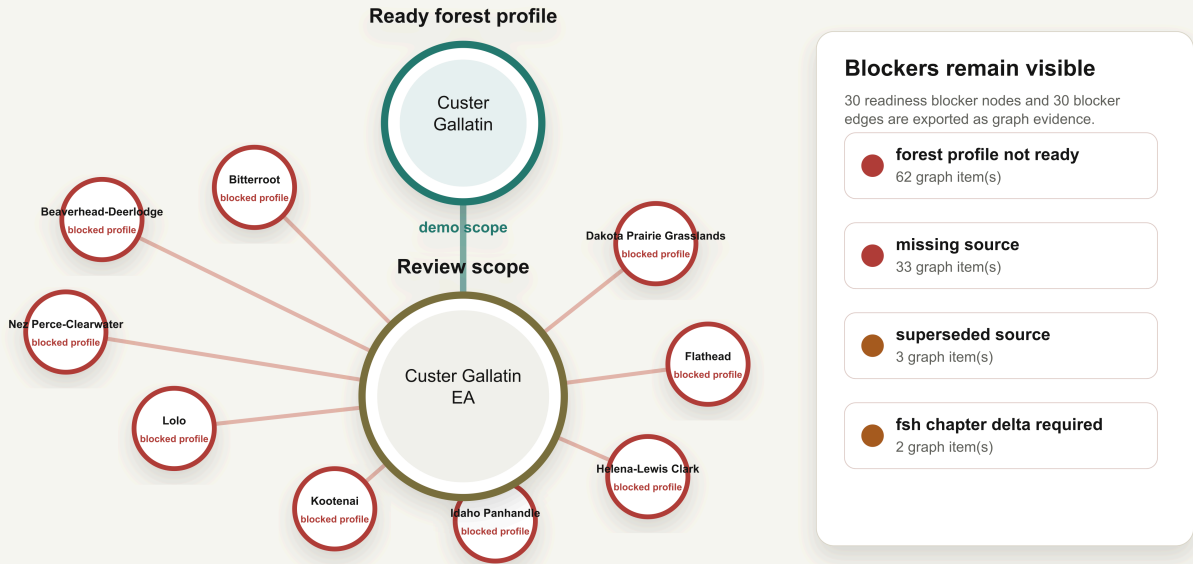
The demo is not just a visualization. It shows that findings can be traced back to the repo evidence model and the validated local graph export.

Show capability without overstating readiness

The demo mode is designed for a client walkthrough: scene buttons move from source library to authority universe, applicability, evidence path, forest plan, readiness blockers, and the full graph. Readiness remains evidence-backed; layout never promotes the review.

Readiness controls prevent overclaiming

The graph can demo a validated Custer Gallatin review while still showing why broader Region 1 coverage is blocked until profile and source evidence is complete.



Demo value

- 1 graph-ready forest profile and 9 blocked forest profiles are visible in one view.
- The viewer can show capability now without implying that every Region 1 profile is ready for compliance promotion.

Readiness scene: blocked profiles and missing-source requirements are graph objects. The system shows Custer Gallatin as demo-ready while preventing broader Region 1 completeness claims.

Source library

Workbook rows, catalog records, artifact links, and provenance.

Authority universe

Currentness, supersession, source records, and authority families.

Applicability

Review-specific applicable and non-applicable decisions.

Evidence path

Clickable source-to-finding trace for defensibility.

Forest Plan

10 forest units, 10 forest plans, and 387 plan components.

Readiness

30 readiness blockers and 30 blocker edges.

What this demonstrates in three minutes

- Authority coverage is explicit and filterable.
- Non-applicable authorities are retained for defensibility.
- Forest-plan scope is visible and blocked when not ready.
- Exports preserve graph state, validation status, and selected evidence paths.

Currentness boundary: CEQ's page records removal of its NEPA regulations and prior rulemakings as rescinded/no longer in effect. Forest Service procedure sources remain an agency-specific implementation layer. The graph visualizes the repo's validated source set and review overlay; it is not a substitute for legal review.